

```

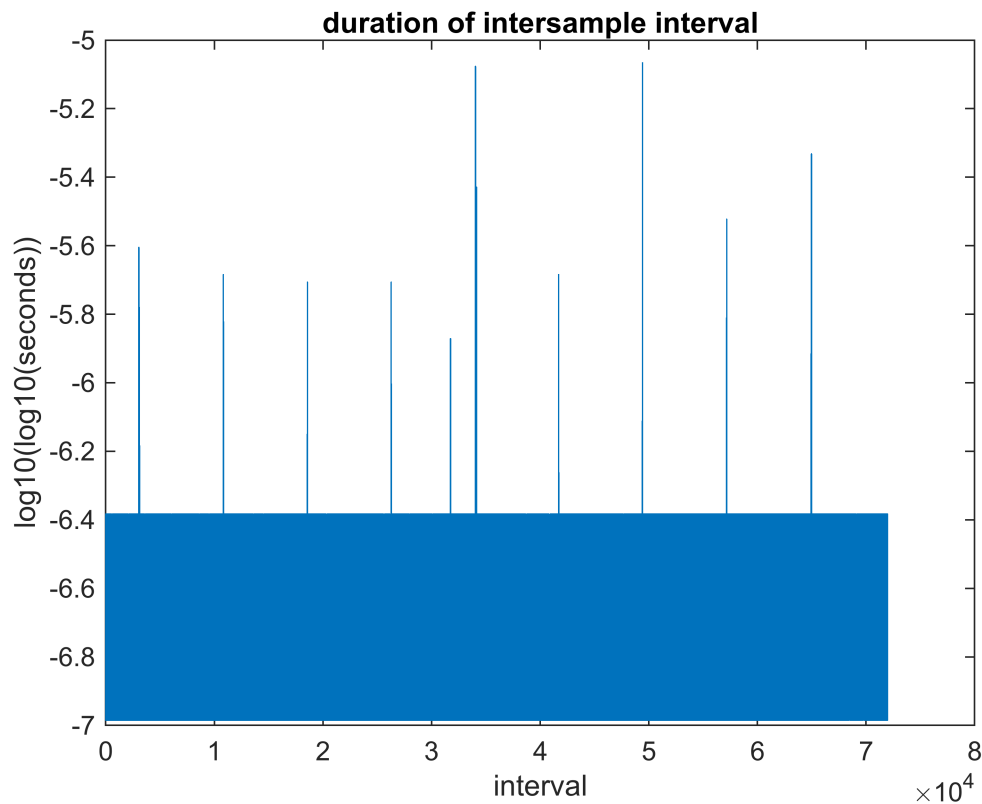
clear
% Import data from text file
%TimeTemp = readtable("\
\apporto.com\dfs\UHCL\Users\1915097_uhcl\Documents\Thesis\TimeTemp.csv");
TimeTemp = readtable("\
\apporto.com\dfs\UHCL\Users\1915097_uhcl\Documents\Thesis\TimeTemp_2026_Sep_06_Sun_1
3_59_53.csv");
%TimeTemp = [TimeTemp;readtable("\
\apporto.com\dfs\UHCL\Users\1915097_uhcl\Documents\Thesis\TimeTemp_2026_Sep_07_Mon_1
0_00_49.csv")]

```

```

% Calculate intersample interval
dt = [1;diff(TimeTemp{:, "now"})];
figure(); plot(log10(log10(dt)))
title("duration of intersample interval")
ylabel("log10(log10(seconds))")
xlabel("interval")

```



```

%compute the Allan Variance of the Intersample Interval Duration
dt_tau = 1:height(dt)/2;
dt_avar=allanvar(dt,dt_tau); % tau channel

%Compute the slope of the clock Allan Variance over Tau
idx = dt_tau >= 1 & dt_tau <= 1000; % sample region

```

```
% linear fit in log10-log10 space;
p = polyfit(log10(dt_tau(idx)), log10(dt_avar(idx)), 1);
disp("LogLog Slope:" + p(1)) % p(1) = slope;
```

LogLog Slope:-0.95167

```
disp("LogLog Intercept:" + p(2)) % p(2) = intercept;
```

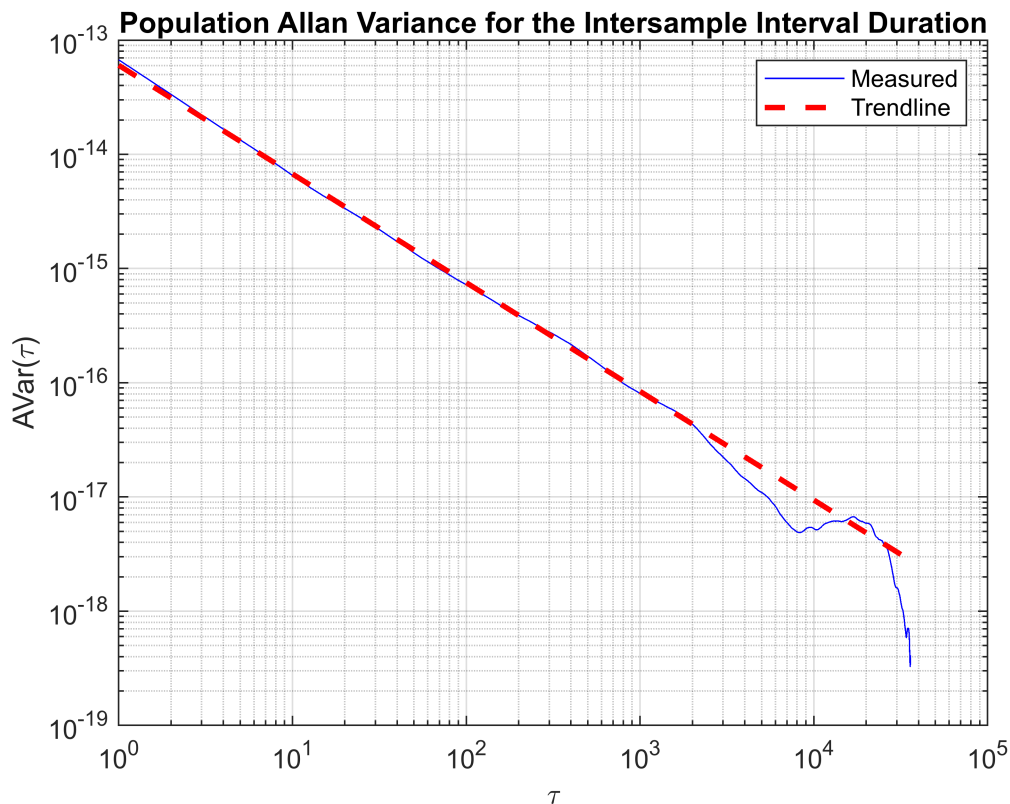
LogLog Intercept:-13.2203

```
disp("AVAR(\tau)=" + p(2) + "\tau^"+ p(1))
```

AVAR(\tau)=-13.2203\tau^-0.95167

```
yfit = 10.^(p(1)*log10(dt_tau) + p(2)); %create the best fit line

figure();
loglog(dt_tau, dt_avar, 'b'); hold on; grid on;
loglog(dt_tau, yfit, 'r--', 'LineWidth', 2);
title("Population Allan Variance for the Intersample Interval Duration");
xlabel('\tau'); ylabel('AVar(\tau)');
legend("Measured", "Trendline");
hold off
```



```
predictionTaus = 10.^[0;1;2;3;4;5;6];
predictedJitter = 10.^(p(1)*log10(predictionTaus) + p(2));
```

```
table(predictionTaus,predictedJitter)
```

```
ans = 7x2 table
```

	predictionTaus	predictedJitter
1	1	6.0209e-14
2	10	6.7296e-15
3	100	7.5218e-16
4	1000	8.4072e-17
5	10000	9.3968e-18
6	100000	1.0503e-18
7	1000000	1.1739e-19

```
% xlim([4.292E4 4.293E4])
% ax = gca;
% chart = ax.Children(1);
% datatip(chart,42923,1.00245);
%
% TimeTemp = TimeTemp(1:42923,:);
```

```
%Create a shortcut to point to the rawdata
```

```
allsensors = ["Temp1","Temp2","Temp3","Temp4","Temp5","Temp6","Temp7","Temp8"];
```

```
%human readable datetime
```

```
%TimeTemp{:,"Days"}=datetime(TimeTemp{1,"now"},"ConvertFrom",'posixtime','timezone',
'America/Chicago');
```

```
TimeTemp{:,"Days"} = TimeTemp{:,"now"}/(60*60*24) + datenum(1970,1,1,-5,0,0);%use a
-5 offset for UCT to CST
```

```
%TimeTemp{:,"DeltaT"}=TimeTemp{:,"now"}-TimeTemp{1,"now"}; %Calculate
```

```
%take the moving average after eliminating outliers from the moving median
```

```
% 0. Compute the Per Row Median
```

```
TimeTemp{:,"median"}=median(TimeTemp{:,allsensors},2);
```

```
% 1. Compute deviation from median
```

```
TimeTemp{:,"MedianDifference"} = abs(TimeTemp{:,allsensors} - TimeTemp{:,"median"});
```

```
% 2. Build mask of good sensors
```

```
TimeTemp{:,"mask"} = TimeTemp{:,"MedianDifference"} <= 0.25; % Nx8 logical
```

```
% 3. Zero out bad sensors
```

```
TimeTemp{:,"goodVals"} = TimeTemp{:,allsensors} .* TimeTemp{:,"mask"};
```

```
% 4. Count good sensors per row
```

```
TimeTemp{:,"numGood"} = sum(TimeTemp{:,"mask"}, 2);
```

```
% 5. Compute robust average
```

```
robustAvg = sum(TimeTemp{:,"goodVals"}, 2) ./ TimeTemp{:,"numGood"};
```

```
% 6. Handle rows with zero good sensors
```

```
robustAvg(TimeTemp{:,"numGood"} == 0) = NaN;
```

```
TimeTemp(:, "robustAvg")=robustAvg;
```

```
clear robustAvg
```

```
% Smooth robustAvg data
```

```
[TimeTemp.robustAvgSmoothGauss,winSize] =  
smoothdata(TimeTemp(:, "robustAvg"), "gaussian", 6000, ...  
    "SamplePoints", TimeTemp.now);
```

```
% Display results
```

```
figure
```

```
plot(TimeTemp.Days, TimeTemp.median, "SeriesIndex", 6, ...  
    "DisplayName", "Input data: robustAvg")
```

```
hold on
```

```
plot(TimeTemp.Days, TimeTemp.robustAvgSmoothGauss, "SeriesIndex", 1, "LineWidth", 1.5,  
    ...  
    "DisplayName", "Smoothed data: robustAvgSmoothGauss")
```

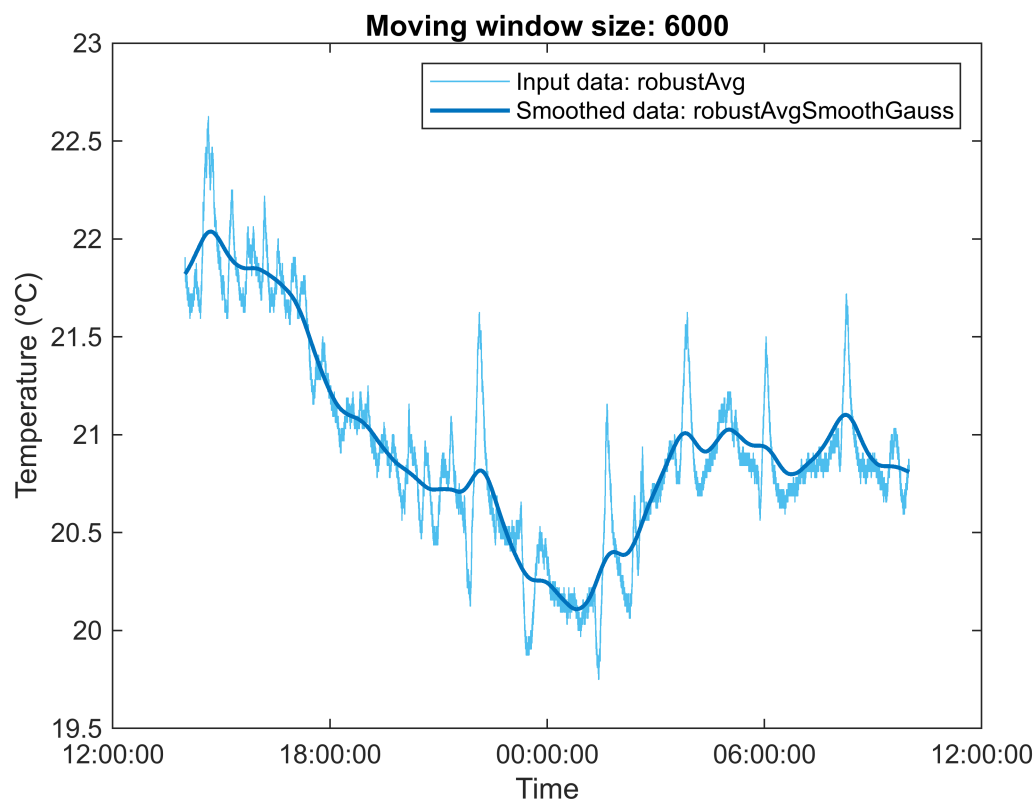
```
hold off
```

```
title("Moving window size: " + string(winSize));
```

```
legend
```

```
ylabel('Temperature (\text{C})')
```

```
xlabel("Time"); datetick('x', 'HH:MM:SS');
```



```
clear winSize
```

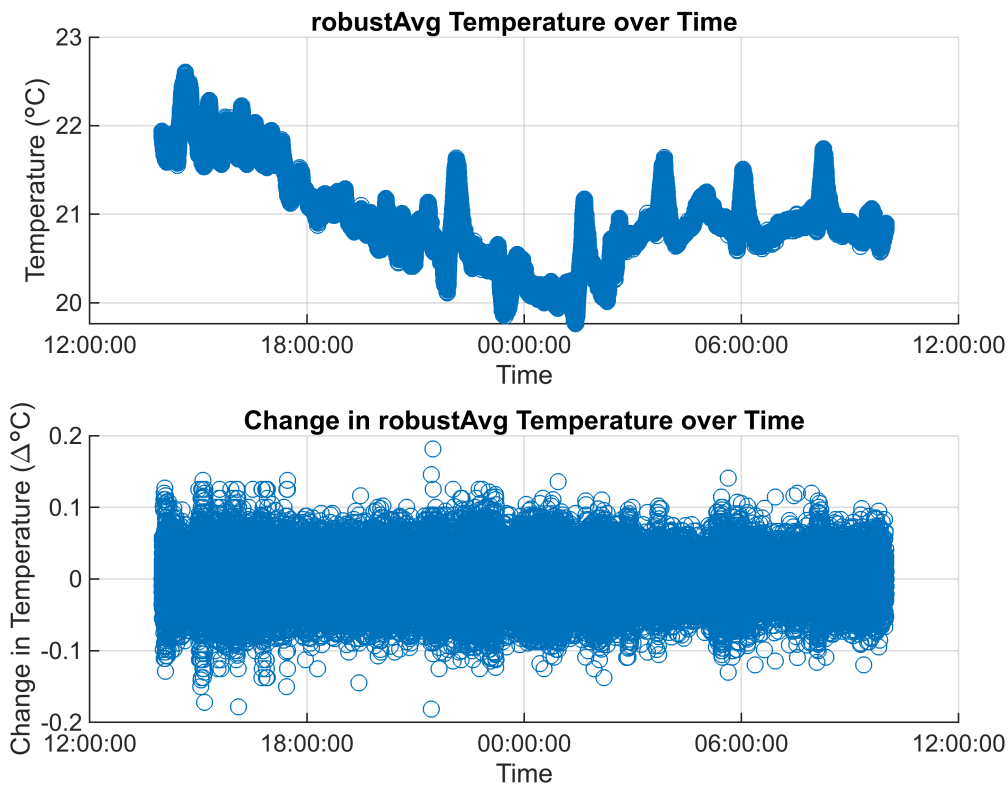
```
%Plot the robustAverage
```

```

figure()
subplot(2,1,1)
scatter(TimeTemp{:, "Days"}, TimeTemp{:, "robustAvg"})
%ylim([21 23.5])
xlabel('Time'); datetick('x', 'HH:MM:SS'); % or 'yyyy-mm-dd HH:MM:SS'
ylabel('Temperature (\text{C})');
title("robustAvg Temperature over Time")
grid on

%Plot the point to point difference
dT = [0; diff(TimeTemp{:, "robustAvg"})];
subplot(2,1,2)
scatter(TimeTemp{:, "Days"}, dT)
%ylim([-1 1]*0.15)
datetick('x', 'HH:MM:SS'); % or 'yyyy-mm-dd HH:MM:SS'
xlabel('Time');
ylabel('Change in Temperature (\Delta\text{C})');
title("Change in robustAvg Temperature over Time")
grid on

```



```

%plot dtime over Temp
%plot dtime over dTemp
figure()

subplot(2,1,1)
scatter(TimeTemp.robustAvg, dt)

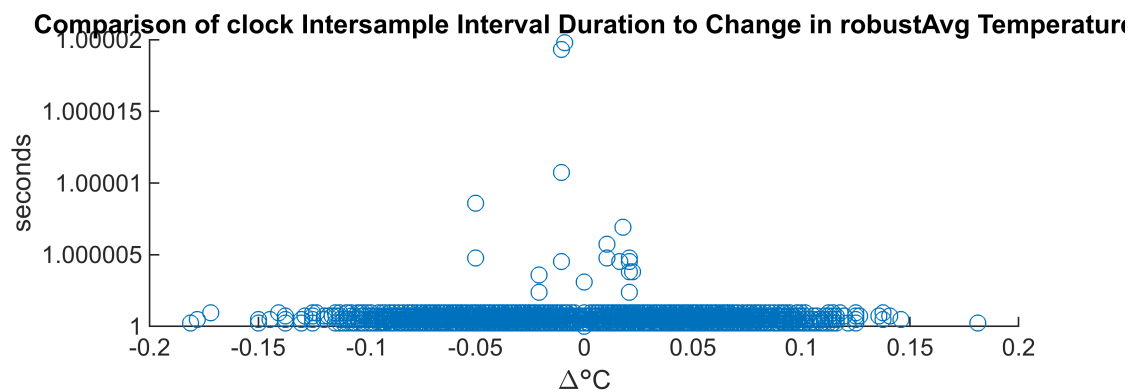
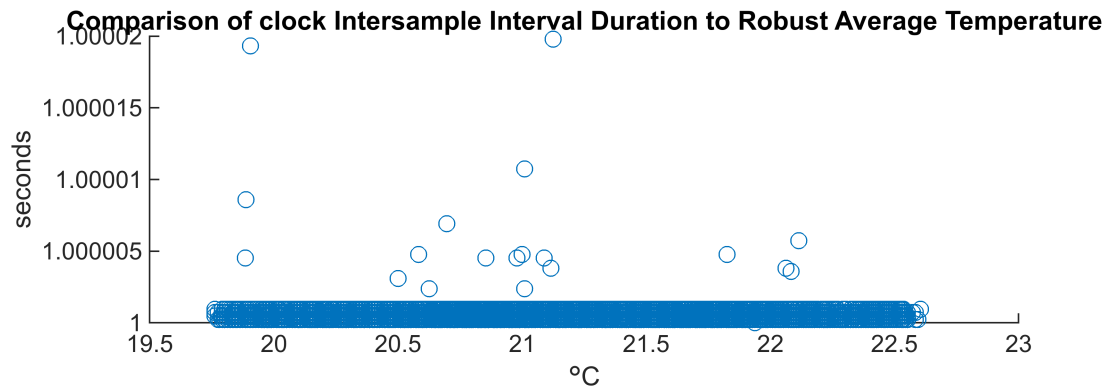
```

```

title("Comparison of clock Intersample Interval Duration to Robust Average
Temperature")
xlabel('\circC');ylabel('seconds');

subplot(2,1,2)
scatter(dT,dt)
title("Comparison of clock Intersample Interval Duration to Change in robustAvg
Temperature")
xlabel('\Delta\circC');ylabel('seconds');

```



```

plotselect = 0;
if plotselect > 0
    figure()
    hold off
    if plotselect == 1 | plotselect == 3
        dotsize = 4;
        scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp1"}, dotsize, "filled")
        hold on
        scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp2"}, dotsize, "filled")
        scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp3"}, dotsize, "filled")
        scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp4"}, dotsize, "filled")
        scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp5"}, dotsize, "filled")
    end
end

```

```

scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp6"}, dotsize, "filled")
scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp7"}, dotsize, "filled")
scatter(TimeTemp{:, "Days"}, TimeTemp{:, "Temp8"}, dotsize, "filled")

end
if plotselect == 2 | plotselect == 3
    scatter(TimeTemp{:, "Days"}, TimeTemp{:, "robustAvg"})
end
hold off
legend()
%ylim([20.5 22.5])
datetick('x', 'HH:MM:SS'); % or 'yyyy-mm-dd HH:MM:SS'
xlabel('Time');
ylabel('Temperature (\circC)');
grid on
end

```

```

channels = [allsensors, "median", "robustAvg", "robustAvgSmoothGauss"]

```

```

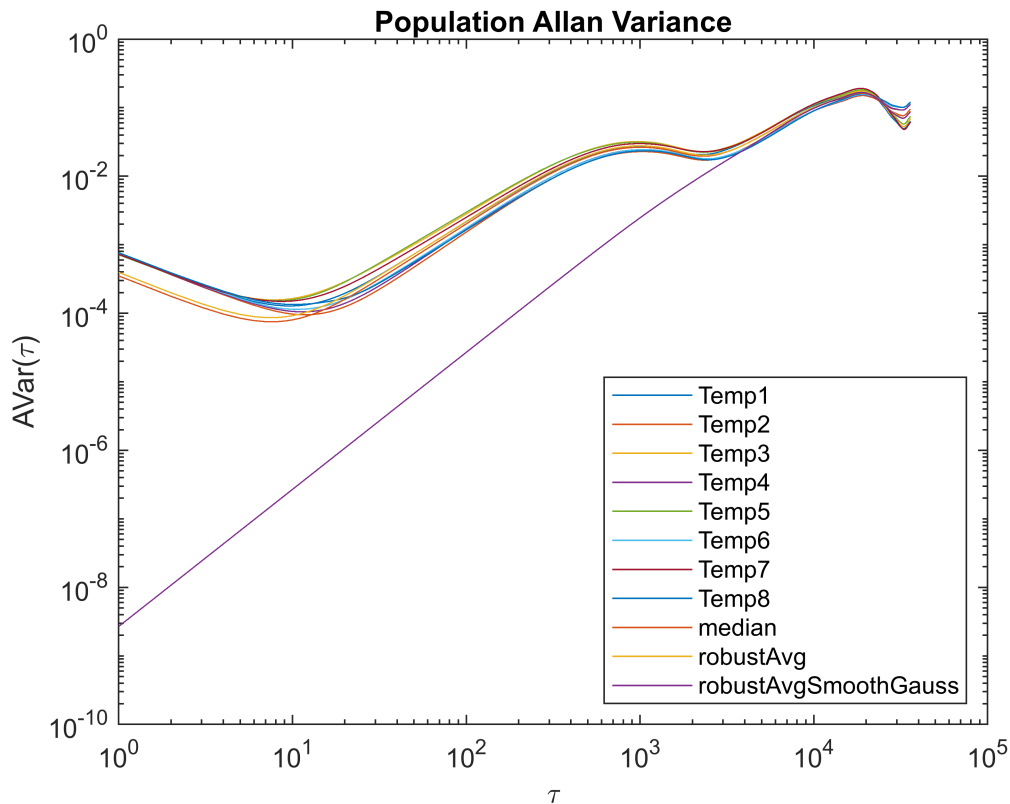
channels = 1x11 string
"Temp1"      "Temp2"      "Temp3"      "Temp4"      "Temp5"      "Temp6"      "T . . ."

```

```

sensorData=TimeTemp{:, channels}; %Samples Sensor
%taufull = unique(round(2.^(0:0.01:log2(size(sensorData,1)/2))));
taufull = 1:height(sensorData)/2;
avarfull=allanvar(sensorData,taufull); % tau channel
figure();loglog(taufull,avarfull);
xlabel('\tau');ylabel('AVar(\tau)');title("Population Allan Variance")
legend([allsensors, "median", "robustAvg", "robustAvgSmoothGauss"], "Location", "southeast")

```



```
%Rolling ALLAN on the data
maxtau=30;minpartitions = 4;
maxpts = maxtau*minpartitions; %seconds of window
%tau = [1,2,4,5,8,10,15,20,%intervals on the binary then to every 5 seconds
tau=[1:maxtau];%unique(round(2.^(0:0.01:log2(maxpts/2)))));
seconds=height(sensorData)-maxpts;
avar = zeros([length(tau) width(sensorData) seconds]);
slices = 1:seconds;

for i = slices
    avar(:, :, i)=allanvar(sensorData(i:i+maxpts,:),tau);
end
size(avar)%avar(tau,channel,slices)
```

```
ans = 1x3
      30      11    71880
```

```
selector=10;
selected_channel = channels(selector)
```

```
selected_channel =
"robustAvg"
```

```
data = squeeze(avar(:,selector,:))';% slices(rows), tau(cols) %select a channel
% %Sort the slices by their minimum tau and amplitude
```

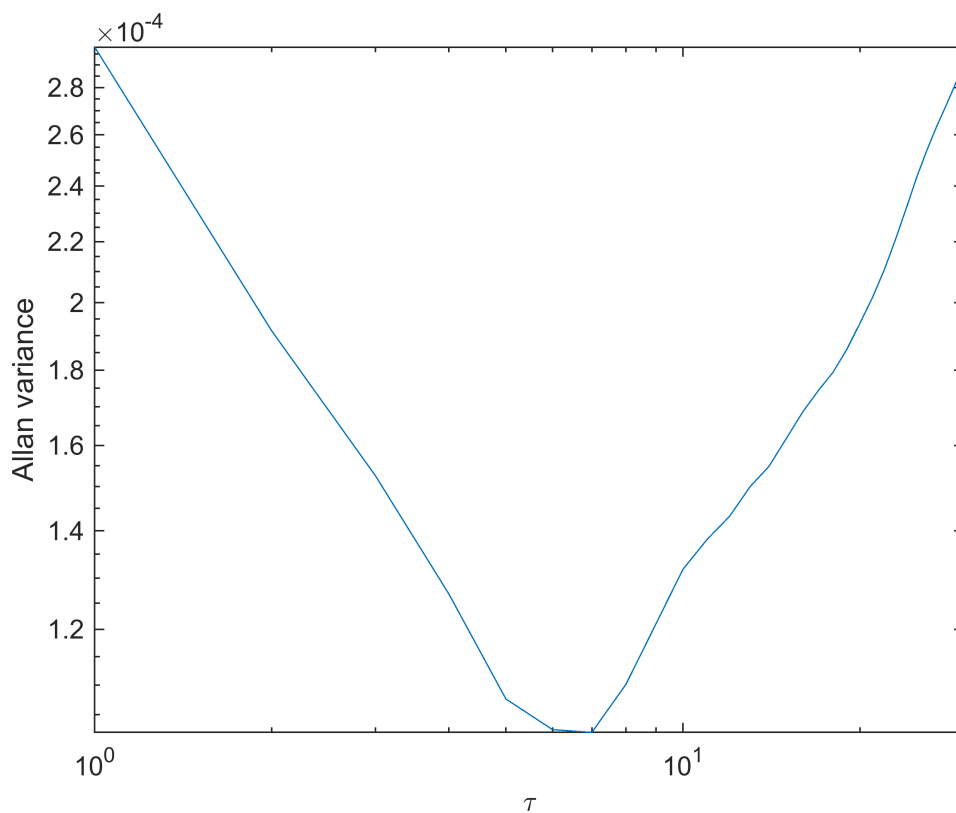
```
[minAvar,minidx]=min(data,[],2);
%[maxAvar,maxidx]=max(data,[],2);

avar_sen=table(data,minAvar,minidx);
clear data minAvar minidx maxAvar maxidx;

% avar_sen.data
% avar_sen.minAvar
% avar_sen.minidx
```

```
%average rolling AVAR Cross Section
figure()
plot(tau, avar_sen.data(1,:))
xlabel('\tau');
ylabel('Allan variance');

%legend("T1","T2","T3","T4","T5","T6","T7","T8","RA","Location","northwest");
set(gca, 'XScale', 'log', 'YScale', 'log');
```

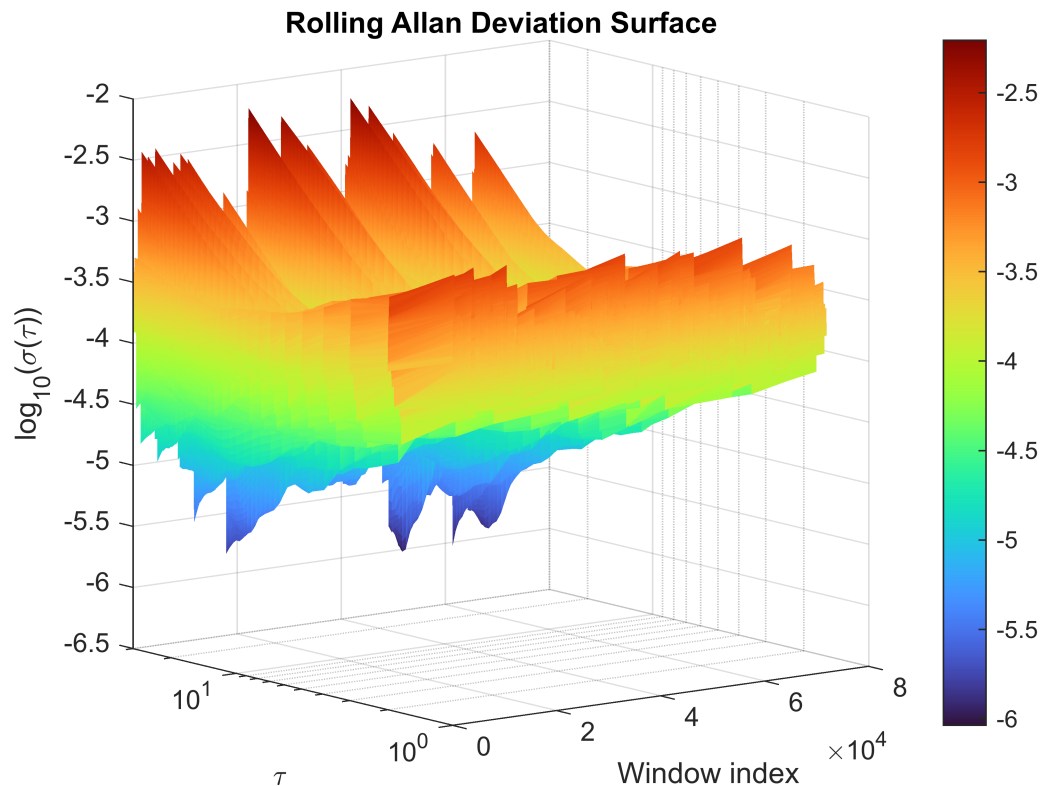


```
%Create a surface with the minimum of each point is marked
figure()
%window = [4.285E+4 42938]
```

```

[TIME,TAU] = meshgrid(slices,tau);
surf(TIME, TAU, log10(avar_sen.data'), 'EdgeColor','none');
hold('on');
%scatter3(slices,tau(avar_sen.minidx),log10(avar_sen.minAvar));
%scatter3(slices,tau(avar_sen.maxAvar),log10(avar_sen.maxAvar));
hold('off');
set(gca,'YScale','log');
xlabel('Window index');ylabel('\tau');zlabel('log_{10}(\sigma(\tau))');
title('Rolling Allan Deviation Surface');
colormap turbo;colorbar;
clim([min(log10(avar_sen.data),[],"all") max(log10(avar_sen.data),[],"all")]);
view(-37.5,30);%Az El

```



```

%xlim(window)
%figure()
%subplot(2,1,1);scatter(slices,log10(avar_sen{:, "minAvar"}));set(gca,'YScale','log')
;xlabel('Window index');ylabel('log_{10}(\sigma(\tau))');%minimum of each point
%subplot(2,1,2);scatter(slices,log10(avar_sen{:, "maxAvar"}));set(gca,'YScale','log')
; xlabel('Window index'); ylabel('log_{10}(\sigma(\tau))');%maximum of each point

% figure()
% subplot(2,1,1)
% scatter(slices,tau(avar_sen.minidx));
% %xlim(window);
% xlabel('index');ylabel('\tau');title('Minimum Rolling Allan Deviation \tau')
%
% subplot(2,1,2)

```

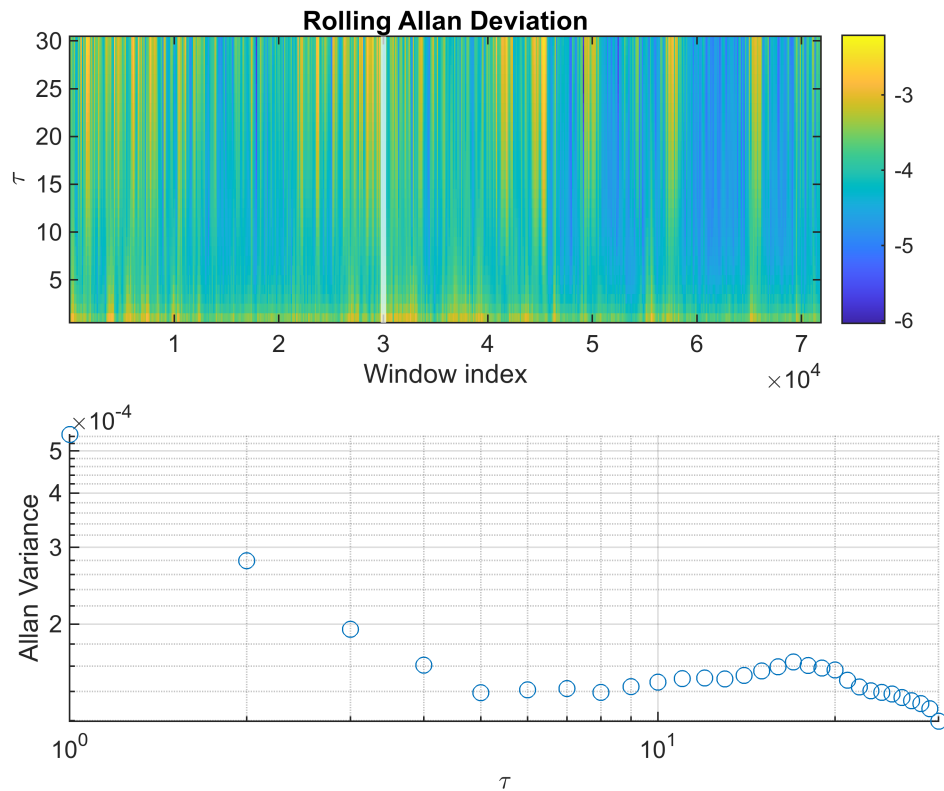
```
% scatter(TimeTemp{:,"Days"},TimeTemp{:,"robustAvg"});
% datetick('x', 'HH:MM:SS'); % or 'yyyy-mm-dd HH:MM:SS'
% xlabel('Time');ylabel('Temperature (°C)');
% grid on
```

```
k = 30000;
figure()
subplot(2,1,1)
imagesc(slices, tau, log10(avar_sen.data));
set(gca,'YDir','normal','YScale','linear');
clim([min(log10(avar_sen.data),[],"all") max(log10(avar_sen.data),[],"all")]);
xlabel('Window index');
ylabel('\tau');
colorbar;
title('Rolling Allan Deviation');

hold on;
%scatter(slices,tau(avar_sen.minidx));
% mask=log10(avar_sen.minAvar)<-25;
% scatter(slices(mask),tau(avar_sen.minidx(mask)));

xline(k, 'w-', 'LineWidth', 2); % white vertical hairline
hold off;

subplot(2,1,2)
scatter(tau, avar_sen.data(k,:));
xlabel('\tau');ylabel('Allan Variance');
set(gca,'XScale','log','YScale','log');
grid('on')
```



```
%Compute the Per Tau FFT
Fs = 1;           % Sampling frequency
T = 1/Fs;         % Sampling period
L = length(slices); % Length of signal
Hz=Fs/L*(0:L-1);
Y_fft = fft(avar_sen.data); %Rows are Time %columns are tau,
%fft: If X is a matrix, then fft(X) treats the columns of X as vectors and returns
the Fourier transform of each column.
```

```
for fft_tau = [1 5 10 15 30]
    mag_fft = abs(Y_fft(:,fft_tau));
    % Find local maxima
    maxIndices =
    islocalmax(mag_fft,"MinSeparation",100,"SamplePoints",Hz,"MaxNumExtrema",1);

    % Display results
    figure();
    plot(Hz,mag_fft,"SeriesIndex",6,"DisplayName","Input data")
    hold on

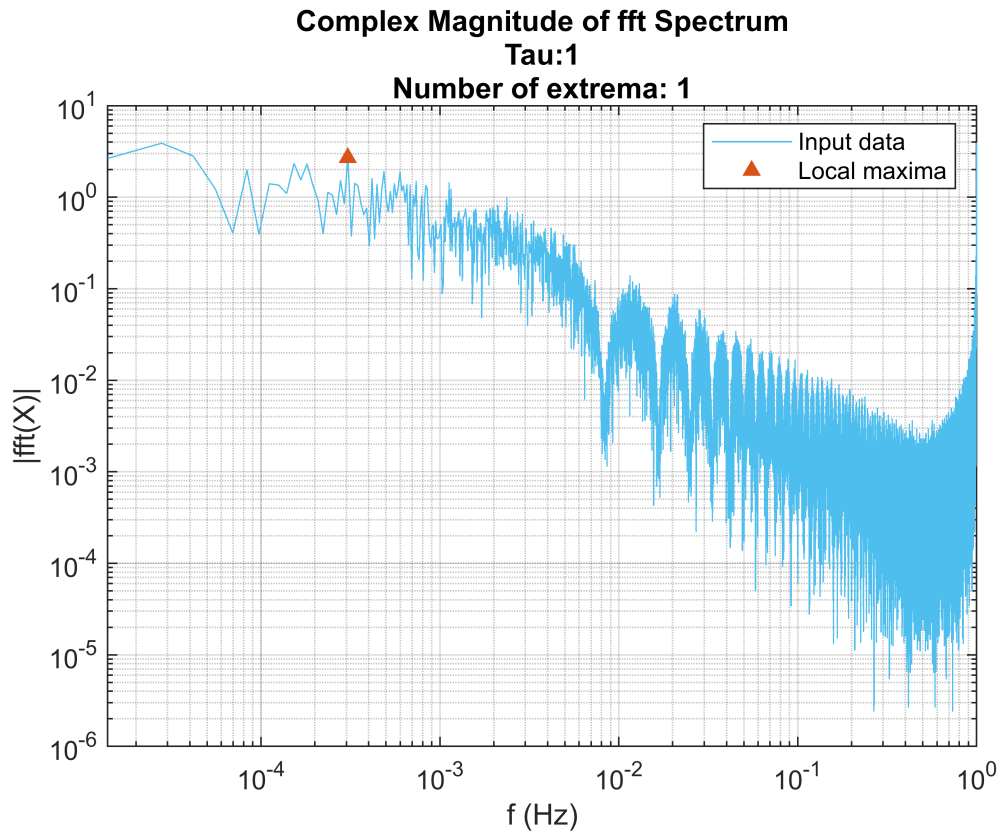
    % Plot local maxima

    scatter(Hz(maxIndices),mag_fft(maxIndices),"^","filled","SeriesIndex",2,"DisplayName",
    "Local maxima")
```

```

    title({"Complex Magnitude of fft Spectrum","Tau:"+fft_tau,"Number of extrema: "
+ nnz(maxIndices)})
    hold off;grid on;
    legend
    xlabel("f (Hz)");ylabel("|fft(X)|");
    set(gca,'XScale','log','YScale','log');
    %xlim([0 0.5]);
end

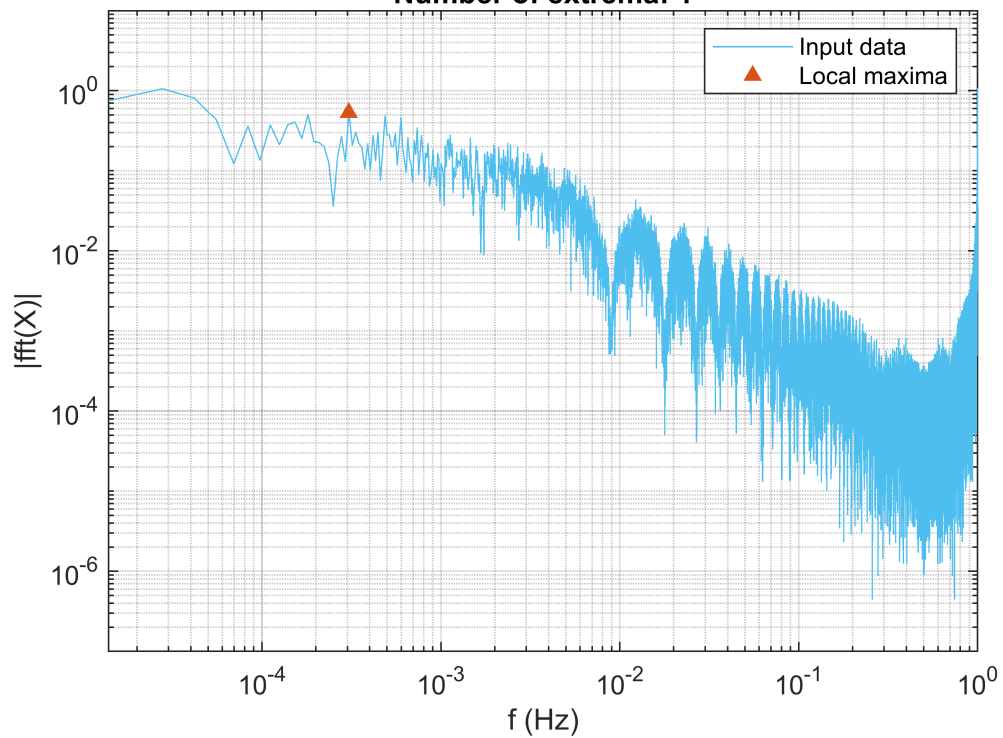
```



Complex Magnitude of fft Spectrum

Tau:5

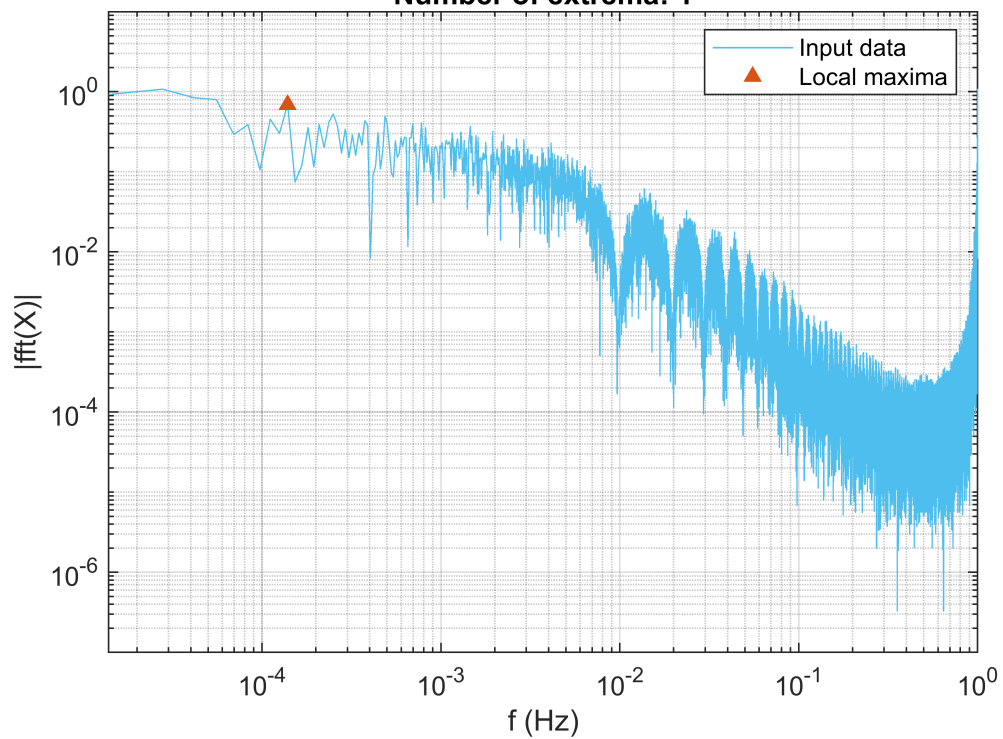
Number of extrema: 1



Complex Magnitude of fft Spectrum

Tau:10

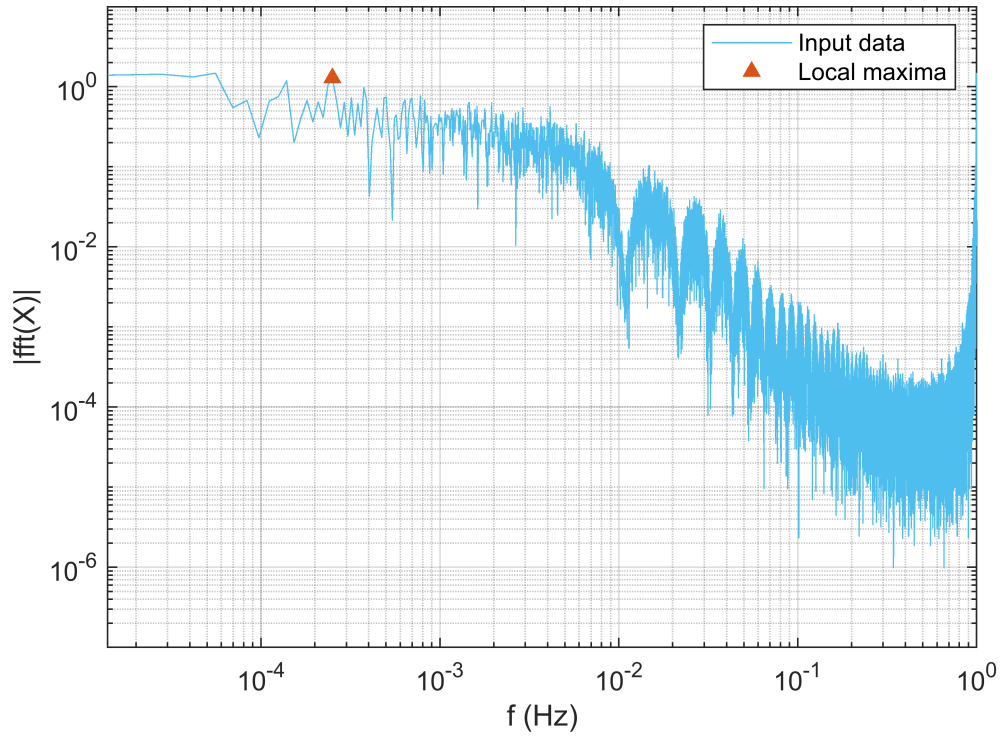
Number of extrema: 1



Complex Magnitude of fft Spectrum

Tau:15

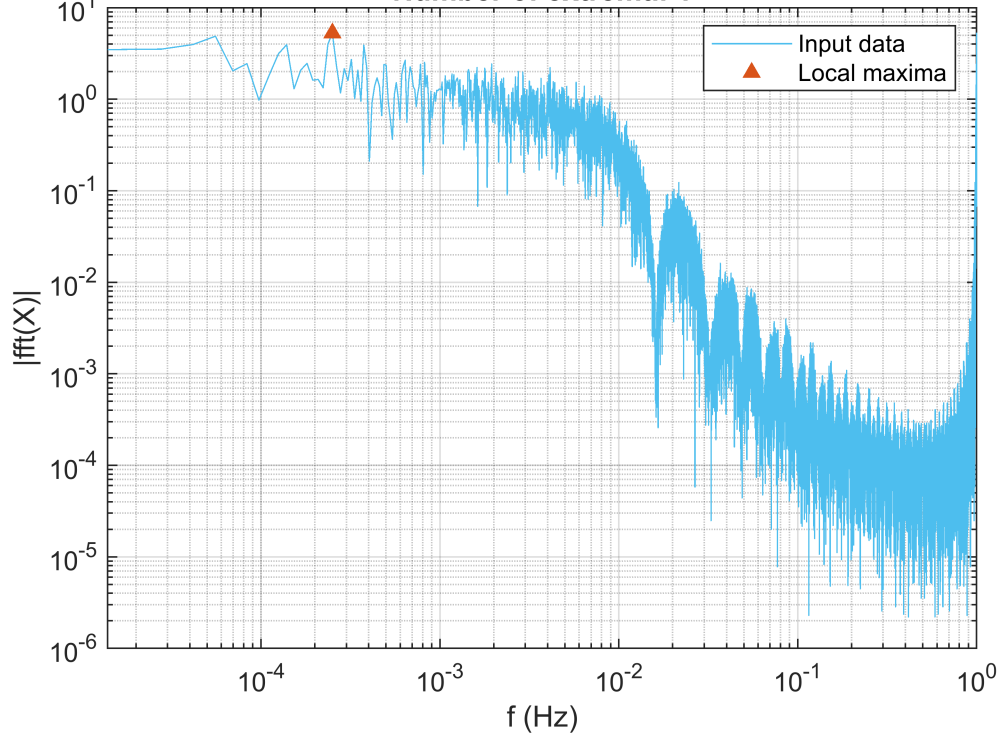
Number of extrema: 1



Complex Magnitude of fft Spectrum

Tau:30

Number of extrema: 1

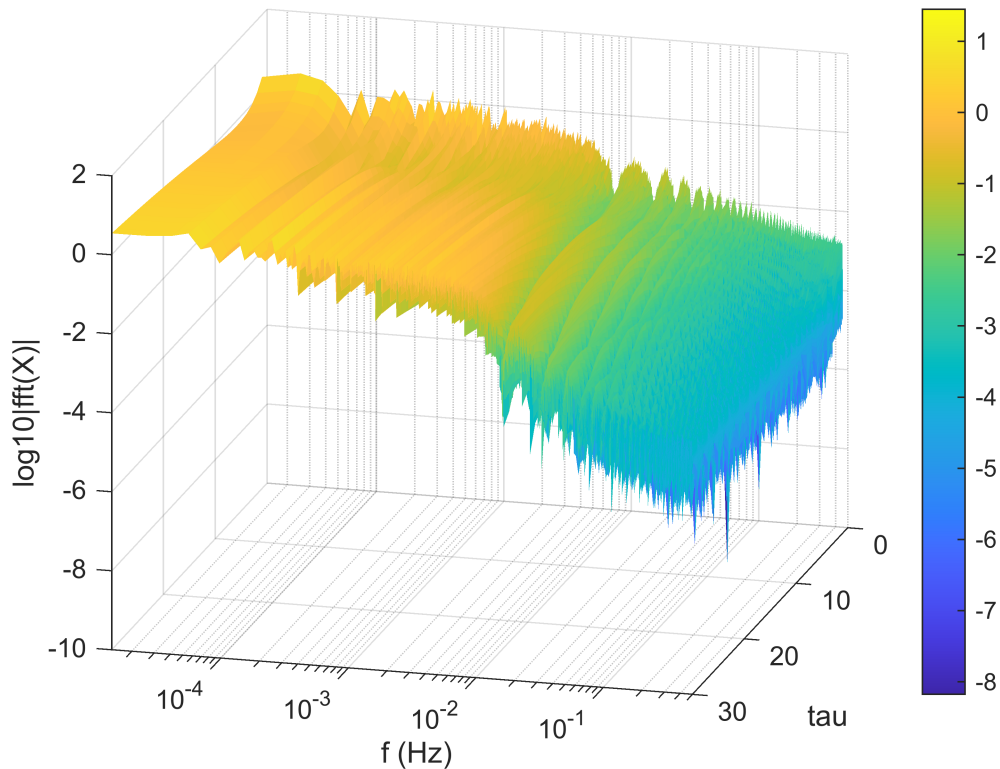


```
figure();
surf(tau,HZ,log10(abs(Y_fft)),"EdgeColor","none");
```

```

xlabel("tau");ylabel("f (Hz)");zlabel("log10|fft(X)|");
set(gca,'XScale','linear','YScale','log','zscale','linear');
view(90+15,20);%Az El
ylim([0 0.5])
colorbar;
%colormap turbo;
colormap parula;

```



%Keeping this because I made it but this is absolutely pointless so far as I can tell

```

%Sort the slices by their minimum tau and amplitude
% avar_sen2 = sortrows(avar_sen,["minidx" "minAvar"]);

```

```

% k = 1000;
% figure()
% subplot(2,1,1)
% imagesc(slices, tau, log10(avar_sen2{:,1:30}'));
% set(gca,'YDir','normal','YScale','log');
% clim([min(log10(avar_sen2{:,1:30}')),[],"all") max(log10(avar_sen2{:,1:30}')),
% [],"all"]]);
% xlabel('Window index');
% ylabel('\tau');
% colorbar;
% title('Rolling Allan Deviation');

```

```

%
% hold on;
% scatter(slices,tau(avar_sen2{:,"minidx"}));
% xline(k, 'w-', 'LineWidth', 2); % white vertical hairline
% hold off;
%
% subplot(2,1,2)
% scatter(tau, avar_sen2{k,1:30});
% xlabel('\tau');ylabel('Allan Variance');
% set(gca,'XScale','log','YScale','log');
% grid('on')

```

```

% figure()
% subplot(2,1,2)
% scatter(slices,tau(avar_sen2{:,"minidx"}));
% %xlim(window);
% xlabel('index');ylabel('\tau');title('Minimum Rolling Allan Deviation \tau')
% subplot(2,1,1)
% surf(TIME, TAU, log10(avar_sen2{:,1:30}'), 'EdgeColor','none');
% hold('on');
% scatter3(slices,tau(avar_sen2{:,"minidx"}),log10(avar_sen2{:,"minAvar"}));
% % scatter3(slices,tau(avar_sen2{:,"maxAvar"}),log10(avar_sen2{:,"maxAvar"}));
% hold('off');
% set(gca,'YScale','log');xlabel('Window index');ylabel('\tau');zlabel('log_{10}
(\sigma(\tau))');title('Rolling Allan Deviation Surface');
% colormap turbo;colorbar;
% view(20,70);

```